Kodimala Sai Veena

Machine Learning Engineer | Software Team Lead

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SUMMARY

Software Team Lead with 5+ years delivering banking solutions at Newgen Software, now moving into Machine Learning. Completed the AI & ML Certificate Program (Masai x IIT Patna) and built ML pipelines and RAG applications with Python, scikit-learn, FastAPI and Docker.

TECHNICAL SKILLS

Machine Learning: Classification, Regression, Clustering, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Evaluation Metrics
GenAI: RAG, Embeddings (sentence-transformers), ChromaDB, LangGraph, Prompt Engineering, Groq LLM API
Programming & Libraries: Python, SQL, Java, NumPy, pandas, scikit-learn, imbalanced-learn, Matplotlib, Seaborn, FastAPI, Pydantic
Tools & Platforms: Git, GitHub, Docker, Jupyter Notebook, JIRA, Newgen iBPS (BPM), OmniDocs, OmniScan, MySQL, Oracle, SQL Server

PROJECTS

Customer Churn Prediction (IBM Telco) | *Python, scikit-learn, pandas* GitHub

- Built a scikit-learn pipeline (encoding, scaling, stratified split) comparing Logistic Regression, Decision Tree and Random Forest.
- Tuned Random Forest via GridSearchCV/RandomizedSearchCV (5-fold CV): recall 46.5% to 79.4%, F1 53.2% to 63.8%.
- Saved the trained pipeline with joblib and built a separate load-and-predict notebook for inference.

Credit Card Fraud Detection | *Python, scikit-learn, imbalanced-learn* GitHub

- Handled 0.17% fraud (284,807 transactions) with class weights, under-sampling and SMOTE; showed SMOTE before splitting leaks data.
- Analysed the precision-recall trade-off (balanced Logistic Regression: recall 91.8%, precision 6.1%) instead of relying on 99.9% accuracy.
- Chose Random Forest by ROC-AUC (97.8% vs 96.7% Logistic Regression + SMOTE, 87.1% Decision Tree); tuned the decision threshold.

Zepto Data & AI Platform - Capstone (graded 85/100) | *Python, SQL, scikit-learn, LangGraph, Docker* GitHub

- Built a data pipeline that scraped paginated data across 3 categories, cleaned it with pandas and queried it with SQL JOINs.
- Titanic survival and fare models: leakage-safe ColumnTransformer, 3 classifiers on F1/ROC-AUC, SMOTE, GridSearchCV, MAE/RMSE/R2.
- Built a RAG support assistant (MiniLM embeddings, ChromaDB, LangGraph routing, Pydantic outputs) served via FastAPI and Docker.

NovaTech RAG Chatbot | *Python, ChromaDB, FastAPI, Groq API, Docker* GitHub

- Chunked documents (100 words, 20 overlap), indexed in ChromaDB and retrieved top-3 chunks with source metadata for each query.
- Prompted a Groq-hosted LLM to answer only from retrieved context and decline unsupported questions; tested out-of-scope queries.
- Exposed the RAG pipeline as a FastAPI service with logging and containerised it with Docker.

Car Price Regression & Customer Segmentation | *Python, scikit-learn* GitHub

- CarDekho price model: Linear Regression test R2 0.74, MAE 212,761 vs 468,748 mean baseline (55% lower); scaling fit on train only.
- Tuned Ridge alpha over 5 iterations; checked train-test gap and 10-fold CV MAE (198,995 to 210,802) to monitor overfitting.
- K-Means customer segmentation: elbow and silhouette analysis (k=2 to 20), chose k=5 and profiled clusters by age, income, spending.

WORK EXPERIENCE

Newgen Software, Chennai Feb 2021 - Present

Team Leader Jul 2024 - Present

- Lead 20+ developers and QA analysts on loan origination and collection platforms for banks; agile practices cut timelines by 50%.
- Built a RAG-based assistant over 5,000-7,000 server/application log records to answer queries like "is anything abnormal?".
- Run requirement gathering, solution design and code reviews, using Git and JIRA to manage delivery.

Senior Software Engineer Jul 2023 - Jun 2024

- Developed banking and financial accounting applications on Newgen iBPS (BPM) from requirements to post-deployment support.
- Wrote Python utilities to load CSV data into the database with business rules; built bulk-upload and email/SMS notification tools.

Software Engineer Feb 2021 - Jun 2023

- Designed and developed Java REST APIs for web and mobile applications.
- Integrated OmniDocs ECM and OmniScan to digitise thousands of daily documents; scanning workflows cut turnaround by 60%+.

EDUCATION

Certificate Program in AI & Machine Learning, Vishlesan i-HUB IIT Patna (Masai) | CGPA 7.91/10 Jan 2026 - Aug 2026

B.E. Computer Science and Engineering, Panimalar Engineering College | CGPA 8.81/10 2017 - 2021

CERTIFICATIONS & AWARDS

Certifications: Tools for Data Science V2, IBM (Coursera, 2023) | Data Science Orientation, IBM (Coursera, 2023) | Business English, Cambridge Assessment English (2019)
Awards: Pinnacle Performance Award, Newgen (MSME digital lending on iBPS, 2025) | Appreciation from City Union Bank (2025) | Black Label Award, Diageo Global (PO Accruals, 2023) | High-Five Award, Newgen (2022)